FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Main)-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A

Sol. Energy density = $\frac{1}{2}\varepsilon_o E_o^2$ and $E_o = (E_{rms})\sqrt{2}$

2. D

Sol. I_L and I_C will be in opposite phases.

$I_{net} = I_L - I_C$

$= 0.6 - 0.4 = 0.2$ A

3. A

Sol. Amplitude and energy both are maximum at resonant frequency.

4. D

Sol. $V = 2(x^2 - y^2)$

Equipotential surfaces will be hyperbolic and $\vec{E}$ everywhere will be $\perp$ to them.

5. A

Sol. $\frac{1.22\,\lambda}{d} = \frac{\Delta r}{x}$

$\Rightarrow \quad x = \frac{(\Delta)\,(d)}{1.22\lambda} = \frac{1\text{mm} \times 3\text{mm}}{1.22 \times 500 \times 10^{-9}\text{m}} = 5$ m

6. B

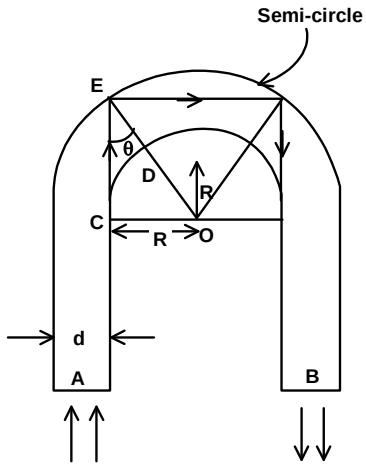
Sol. Source and observer are receding with a constant relative velocity.

7. A

Sol. Above curie temperature ferromagnetic substance become paramagnetic.

8. A

Sol. As shown in the figure, angle of incidence will be minimum for the ray travelling along the inner edge. So the whole incident light entering the glass through surface A will emerge from the glass through surface B.



$\theta > I_C$

$\sin\theta > \sin I_C$

But as, $\sin I_C = \frac{1}{\mu} = \frac{2}{3}$

And from the geometry of the figure,

$$\sin\theta = \frac{OC}{OE} = \frac{R}{OD + DE} = \frac{R}{R + d}$$

so, $\frac{R}{R+d} > \frac{2}{3}$, i.e. $R > 2d$

i.e., $\frac{d}{R} < \frac{1}{2}$ so $\left(\frac{d}{R}\right) = \frac{1}{2}$

9. B

Sol. Total momentum = $\int_{\theta=-90}^{+90} (dm)v\cos\theta$

$dm = \frac{M}{\pi R} \times R d\theta$

Total momentum = $\int \frac{M}{\pi R} v R d\theta \cos\theta = \frac{2Mv}{\pi}$

10. C

Sol. When centre of cube is directly above the edge of cube.

$\Rightarrow \theta = 45°$.

11. C

Sol. For charging $q = CE(1 - e^{-t/RC})$

Charge at $t = RC \Rightarrow q_o = CE(1 - e^{-1})$

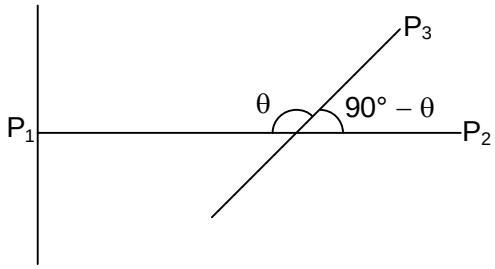
At t = RC discharging starts

$\Rightarrow q = q_o(e^{-t/RC}) = CE(1 - e^{-1}) \times \frac{1}{e} = CE\left(\frac{1}{e} - \frac{1}{e^2}\right)$

12. A

Sol. No light is emitted from the second polaroid, so P_1 and P_2 are perpendicular to each other

Let the initial intensity of light is I_0. So Intensity of light after transmission from first polaroid $= \frac{I_0}{2}$.



Intensity of light emitted from P_3

$$I_1 = \frac{I_0}{2}\cos^2\theta$$

Intensity of light transmitted from polaroid i.e., from

$$P_2 = I_1\cos^2(90^\circ - \theta) = \frac{I_0}{2}\cos^2\theta\sin^2\theta$$

$$P_2 = \frac{I_0}{8}(2\sin\theta\cos\theta)^2 = \frac{I_0}{8}\sin^2 2\theta$$

13. B

Sol. $dV = -IdR = = -I\rho\frac{dx}{A}$

$$\frac{dV}{dx} = -I\frac{\rho}{A} = -k$$

$\Rightarrow R = k\ell/I$

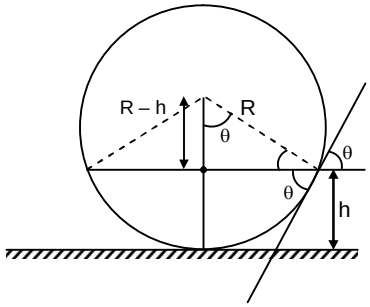
II Method:

$$V_2 - V_1 = \int\left(\frac{dV}{d\ell}\right)d\ell = kL = IR$$

14. B

Sol. $\cos\theta = \frac{R-h}{R}$

$\Rightarrow \cos^{-1}(\theta) = \frac{R-h}{R}$



15. D

Sol. No heat will be produced as no charge flows through S_2 when it is closed.

16. A

Sol. Total resistance = $\frac{500}{3} + \frac{\left(100+\frac{100^2}{25}\right)\left(600+\frac{200^2}{100}\right)}{100+\frac{100^2}{25}+600+\frac{200^2}{100}} = 500$

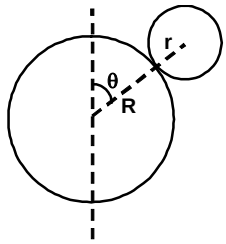
Required heat = $\frac{200^2}{500}$ = 80 J.

17. A

Sol. When the line joining the centers of the two spheres makes an angle θ with the vertical, then

$$mg\cos\theta - N = \frac{mv^2}{R+r}$$

When the contact breaks off, N= 0



$mg\cos\theta = \frac{mv^2}{R+r} = \frac{m\omega^2r^2}{R+r}$ . . . (1) [$\because v = \omega r$]

From COE principle

$mg(R + r) = mg\,(R+r)\cos\theta + \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$

$mg(R + r) = (R+r)\frac{m\omega^2r^2}{R+r} + \frac{m\omega^2r^2}{2} + \frac{1}{2}.\ \frac{2}{5}mr^2\omega^2$

or $g(R + r) = \frac{17}{10}\omega^2r^2$

$\therefore\ \omega = \sqrt{\frac{10g(R+r)}{17r^2}}$

18. C

Sol. For closed organ pipe $f_C = (2n-1)\frac{v}{4L_C}$ with n = 1, 2, …

While for open organ pipe $f_O = n\frac{v}{2L_O}$ with n = 1, 2, …

According to given problem

$f_C = \frac{v}{4L_C} = 110$ i.e., $L_C = \frac{330}{4\times110} = 0.75$ m

Furthermore

$\frac{3v}{4L_C} \sim \frac{2v}{2L_O} = 2.2$ or $\frac{330}{L_O} = 330 \mp 2.2$ $\left(\text{as } \frac{3v}{4L_C} = 3f_C = 330\right)$

i.e., $L_O = \frac{330}{330 \mp 2.2} \Rightarrow L_O = 1.00067$ m or 0.9937 m

19. C

Sol. Maximum velocity $v_{max} = \sqrt{2as}$

$v_{avg} = \frac{6s}{T}$ $T = t_1 + t_2 + t_3$

$t_1 = \sqrt{\frac{2s}{a}},$ $t_2 = \frac{2s}{v_{max}} = \sqrt{\frac{2s}{a}}$

Retardation $\Rightarrow$ B

$\beta = \frac{v_{max}^2}{6s}$

$\beta = \frac{2a\times s}{6\times s} = \frac{a}{3}$

$t_3 = \frac{v_{max}}{\beta} = 3\sqrt{\frac{2a\times s}{a}} = 3\sqrt{\frac{2s}{a}}$

$T = 5\sqrt{\frac{2s}{a}}$

$$v_{avg} = \frac{6}{5}\frac{5}{\sqrt{\frac{2s}{a}}}$$

$$v_{avg} = \frac{6}{5}\sqrt{\frac{as}{2}}$$

$$\frac{v_{avg}}{v_{max}} = \frac{6}{5}\sqrt{\frac{as}{2}} \times \frac{1}{\sqrt{2as}} = \frac{3}{5}$$

20. A

Sol. $\vec{v}_{avg} = \frac{\int_0^{4/5} \vec{v}dt}{\int_0^{4/5} dt}$

21. A

Sol. The resistivity of pure silicon is 2300 Ω m
and μ_e = 0.135 m^2/v-s, μ_h = 0.048 m^2/v-s.
Using
$\sigma = 1/\rho = (n_I \mu_e + n_I \mu_h)e$
$(2300)^{-1} = n_I (0.135 + 0.048) \times 1.6 \times 10^{-19}$
$n_I = 1.5 \times 10^{16}$ /m^3.
Is the intrinsic electron & hole concentration. The resistivity of a specimen doped with 10^{19} P-atoms/m^3 can be found from :
σ (conductivity) = n_e e.μ_e ($\because n_e = 10^{19}/m^3 >> n_I$)
$= 10^{19} \times 1.6 \times 10^{-19} \times 0.135 = 0.216$ mho / m.

$\rho = \frac{1}{\sigma}$ = 4.6 Ω -m

22. B

Sol. Power = 10W

$\Rightarrow$ Force = $\frac{P}{C}$ (for perfectly absorbing surface)

$F = \frac{10}{3\times10^8 m/s} = \frac{1}{3}\times10^{-7}$

23. B

Sol. $v_{rms} = \sqrt{\frac{3RT}{M}}$

24. C

Sol. $E_g(eV) = \frac{1242}{\lambda(nm)} = \frac{1242}{620} = 2eV$

25. C

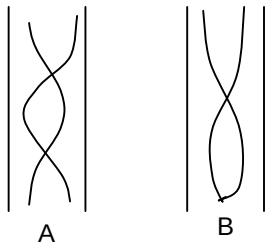
Sol. (a) $2\frac{V_A}{2L_A} = 3\frac{V_B}{4L_B}$

$\Rightarrow \frac{V_A}{V_B} = \frac{3}{4}$ $\left[\because L_A = L_B\right]$

$V_A = \sqrt{\frac{\gamma_A RT}{M_A}}, \quad V_B = \sqrt{\frac{\gamma_B RT}{M_B}}$

$\Rightarrow \sqrt{\frac{\gamma_A}{\gamma_B}\frac{M_B}{M_A}} = \frac{3}{4}, \quad \gamma_A = \frac{5}{3}, \quad \gamma_B = \frac{7}{5}$

$\Rightarrow \frac{M_A}{M_B} = \frac{400}{189}$



26. A

Sol. Given $\eta_1 = \frac{1}{6}$, $\eta_2 = \frac{1}{3}$

If the temperatures of the source and the sink between which the cycle is working are T_1 and T_2, then the efficiency in the first case will be

$\eta_1 = 1 - \frac{T_2}{T_1} = \frac{1}{6}$

In the second case $\eta_2 = 1 - \frac{T_2 - 65}{T_1} = \frac{1}{3}$

Solving $T_1 = 390$ K and $T_2 = 325$ K.

27. B

Sol. $\Delta V = \frac{B\omega^2(\ell)}{2} = \frac{B\omega^2(\sqrt{3}\,R)}{2} = V_B - V_A.$

28. B

Sol. At steady state.

Rate of generation of A = Rate of decay of A.

$q = \lambda N_A$

$\Rightarrow N_A = \frac{q}{\lambda}$

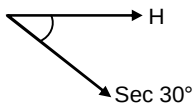
29. B

Sol. For series combination $R = 220^2\left(\frac{1}{200} + \frac{1}{100}\right) = 220^2 \times \frac{3}{200}$

$\therefore$ Power $= \frac{220^2}{R} = \frac{200}{3}$ W

30. B

Sol.



$\therefore$ H sec 30° = $2H/\sqrt{3}$

Chemistry

PART – II

SECTION – A

31. C
Sol. I.E = $-E_1$ = - (-13.6) = 13.6 eV.

32. C
Sol. $\Delta n = n_P - n_R = 1 - 1 = 0$
$\therefore$ Pressure has no effect.

33. C
Sol. $^{3}_{1}T \longrightarrow {}^{3}_{2}He + {}^{0}_{-1}e$

34. B
Sol. Antipyretics reduce body temperature.

35. D
Sol. Hybridization of boron changes to sp^3.

36. C
Sol. Both NaCl and KNO_3 are neutral salt.

37. C
Sol. KH is not formed as there is not H_2 in the atmosphere.

38. B
Sol. For first order reaction, $t_{1/2} = \frac{0.693}{k}$

39. B
Sol. Electrophilic substitution reaction takes place.

40. A
Sol. The products are CH_3OH, CH_3I, C_2H_5OH and C_2H_5I.

41. A
Sol. Due to strong B – F bond energy.

42. C
Sol. d- or ℓ- and diastereomers rotate plane polarized light.

43. B
Sol. Expansion of ring takes place in (B). In(C) the initially formed carbocation does not undergo rearrangement to initiate ring expansion.

44. B
Sol. Cleavage for expoxy ring takes place in acidic medium.

45. B

Sol. $HCHO + CH_3MgBr \xrightarrow{H_2O/H^+} CH_3CH_2OH$

46. C

Sol. It contains no chiral atom, $H_2N - CH_2 - COOH$.

47. B

Sol. Phenol forms complex with $FeCl_3$ and changes its colour.

48. B

Sol. It is a polymer of $CF_2 = CF_2$.

49. C

Sol. n-factor of NH_4NO_2 is 3.

50. B

Sol. Maximum number of molecules of a gas moves with most probable velocity.

51. A

Sol. On moving down the group, the s-character of the orbital that holds the lone pair increases. Hence basicity decreases.

52. A

Sol. $C + 2H_2SO_4 \longrightarrow CO_2 + 2SO_2 + 2H_2O$

53. D

Sol. In Cl_2O_7, it's oxidation number is maximum.

54. A

Sol. Oxygen forms both sigma and pi bonds with xenon.

55. C

Sol. Esters are not reduced by $NaBH4_4$.

56. B

Sol. Br_2/H_2O is a mild oxidizing agent.

57. B

Sol. $SO_2 + 2H_2S \longrightarrow 3S + 2H_2O$

58. C

Sol. Br_2 exists as a liquid at room temperature.

59. A

Sol. $E = E^0 - \frac{0.0591}{2}\log\left[H^+\right]^2, \left[H^+\right] = 10^{-3}, E^0 = 1.30\,V$

60. C

Sol. No. of tetrahedral voids = 12
No. of octahedral voids = 6

Mathematics

PART – III

SECTION – A

61. A

Sol. Clearly two sides are $x-2y+1=0$ and $x+y=0$

As orthocenter lies on $x-2y+1=0$ is right angled (let at B)

Third side passes through (1, 1) and is perpendicular to $x-2y+1=0$

$\Rightarrow$ Third side is $2x+y-3=0$

$\therefore$ Circum – radius $=\frac{1}{2}AC=\frac{5\sqrt{2}}{3}$

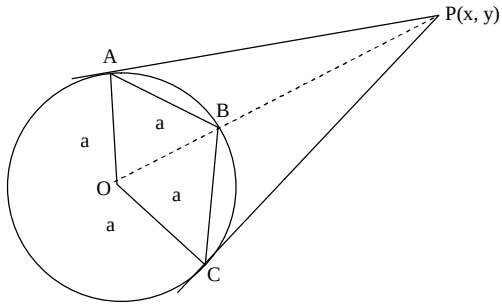
62. B

Sol. $AB=AO=BO=a\Rightarrow\angle AOB=60^\circ$

$\Rightarrow\frac{PA}{r}=\sqrt{3}$

$\Rightarrow x^2+y^2-a^2=3a^2$

$\Rightarrow x^2+y^2=4a^2$



63. D

Sol. $x^2+9y^2-4x+6y+4=0$

$\Rightarrow \quad (x-2)^2+9\left(y^2+\frac{2}{3}y+\frac{1}{9}\right)=1$

$\Rightarrow \quad \frac{(x-2)^2}{1}+\frac{\left(y+\frac{1}{3}\right)^2}{\left(\frac{1}{9}\right)}=1$

Its parametric equation is

$x=2+\cos\theta, y=-\frac{1}{3}+\frac{1}{3}\sin\theta$

Hence $4x-9y=8+4\cos\theta+3-3\sin\theta$

$\Rightarrow$ minimum value of $4x-9y$ is 6

64. A

Sol. Equation of hyperbola $x^2+y^2=(x+y+1)^2$

$2xy+2x+2y+1=0$

$\therefore$ Equation of Asymptotes $2xy+2x+2y+1+\lambda=0$. For a pair of straight lines $\lambda=1$

$\therefore$ Asymptoes : $(y+1)(x+1)=0$ or $y=-1$ and $x=-1$

65. B

Sol. Length $= \frac{2ab}{a^2m^2+b^2}\sqrt{\left(1+m^2\right)\left(a^2m^2+b^2+c^2\right)}$ where $m=1, c^2=(ae)^2$

$$= \frac{2\left(\frac{2b^2}{a}\right)}{1+\frac{b^2}{a^2}} = \frac{2\left(\frac{2b^2}{a}\right)}{1+1-e^2} = 6.\frac{6}{7}$$

66. C

Sol. Equation of normal at $(2\sec\theta, \tan\theta)$ is $\frac{2x}{\sec\theta}+\frac{y}{\tan\theta}=5$

Slope $=-1 \Rightarrow -\frac{2\tan\theta}{\sec\theta}=-1$

$\Rightarrow \quad \theta=\frac{\pi}{6}$

$\therefore$ normal becomes $y=-x+\frac{5}{\sqrt{3}}$

$\because$ It is tangent to ellipse $\Rightarrow a^2+b^2=\frac{25}{3} \Rightarrow \frac{9}{25}\left(a^2+b^2\right)=3$

67. C

Sol Let α be a root of equation $x^2-3x+c=0$ then $-\alpha$ is a root of equation

$x^2+3x-c=0 \Rightarrow c=0$

$\Rightarrow$ the smallest of $x^2-3x=0$ are 0 and 3

68. B

Sol. $x=1+2+3+.....49$

$$=\sum_{n=1}^{49} n$$

$$=\frac{n(n+1)}{2}, \text{ where } n=49 \qquad \text{(i)}$$

Let $1^3+2^3+3^3+.....+49^3=y$

$$y=\sum_{n=1}^{49} n^3=\left(\frac{n(n+1)}{2}\right)^2, \text{ where n = 49} \qquad \text{(ii)}$$

Hence $y=x^2$ by comparing (i) & (ii)

69. D

Sol. Let $z=e^{ix}\ x\in[0, 2\pi)$

$$\left|\frac{z}{\bar{z}}+\frac{\bar{z}}{z}\right|=1 \Rightarrow \left|z^2+\bar{z}^2\right|=1$$

$$\left|\cos 2x+i\sin 2x+\cos 2x-i\sin 2x\right|=1$$

$\Rightarrow \cos 2x = \pm\frac{1}{2}$

$x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}\left(\cos x = \frac{1}{2}\right)$

$x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}\left(\cos x = \frac{-1}{2}\right)$

70. C

Sol. $(1+x)^{10}(2+x)^{11}(3+x)^{12}$

$= \left(x^{10} + {}^{10}C_1x^9 + ...\right)\left(x^{11} + {}^{11}C_1x^{10}(2) + ...\right)\left(x^{12} + {}^{12}C_1x^{11}(3) + ...\right)$

coefficient of $x^{32} = (12)(3) + (11)(2) = (10) = 68$

71. A

Sol. $\vec{a}.\vec{b} = 0 \Rightarrow x_1 + x_2 + x_3 = 0$

Thus we have to obtain the number of integral solution of this equation.

Coefficient of $x^\circ \mid (x^{-3} + x^{-2} + x^{-1} + x^0 + x + x^2)^3$

$= x^\circ \left|\left(\frac{1+x+x^2+x^3+x^4+x^5}{x^3}\right)^3\right.$

$= x^9 \mid (1-x^6)^3 (1-x)^{-3} = {}^{11}C_9 - 3.{}^5C_3 = 25$.

72. B

Sol. $\cos 5\theta = \cos 8\theta$

$\cos 3\theta = \cos 16\theta$

$\therefore \cos\theta \cos 2\theta \cos 4\theta \cos 8\theta \cos 16\theta \cos 32\theta$

$= \frac{\sin 64\theta}{2^6 \sin\theta} = -\frac{\sin\theta}{2^6 \sin\theta} = \frac{-1}{2^6} = P$

$2^8|P| = 2^8 \times \frac{1}{2^6} = 4$

73. C

Sol. $\cos 60^o = \frac{b^2c^2 - a^2}{2bc} \Rightarrow bc = a^2$

$b + c = 2a \Rightarrow b - c = 0$

$\therefore$ Triangle is equation

$A\ \frac{\sqrt{3}}{4}a^2 = \frac{1}{2}(3+4+5)a$

$a = \frac{24}{\sqrt{3}}$

$R = \frac{abc}{4} = \frac{a}{\sqrt{3}}$

74. C

Sol. $T_n = \tan^{-1}\left(\frac{(k^2+k+1)-(k^2-k+1)}{1+(k^2+k+1)(k^2-k+1)}\right)$

$= \tan^{-1}(k^2+k+1) - \tan^{-1}(k^2-k+1)$

$S_n = \tan^{-1}(n^2+n+1) - \tan^{-1}(1)$

$= \tan^{-1}\left(\frac{n^2+n+1-1}{1+n^2+n+1}\right) = \tan^{-1}\left(\frac{n^2+n}{2+n+n^2}\right) = \tan^{-1}\left(\frac{6}{7}\right) \Rightarrow n = 3$

$\therefore 2n = 6$

75. A

Sol. $\frac{\sin(xy)}{\cos(xy)} = y$

$\Rightarrow \sin(xy) = xy$

$\Rightarrow xy = 0$

$\Rightarrow x = 0$ or $y = 0$

But $x = 0$ is not possible

$\therefore$ $y = 0$ and $x = 1$, i.e. (1, 0)

76. B

Sol. $g(x) = \frac{x^3 - x^2 - 1}{x(x-1)} - x + 1 = \frac{-2 + 2x^2 - 2x}{2x(x-1)} = \frac{x^2 - x - 1}{x(x-1)}$

Now for $y = \sqrt{g(x)}$, we must have $\frac{x^2 - x - 1}{x(x-1)} \geq 0$ or $\frac{\left(x - \frac{1-\sqrt{5}}{2}\right)\left(x - \frac{1+\sqrt{5}}{2}\right)}{x(x-1)} \geq 0$

$\Rightarrow x \in \left(x \in -\infty, \frac{1-\sqrt{5}}{2}\right] \cup (0, 1) \cup \left[\frac{1+\sqrt{5}}{2}, \infty\right)$

77. C

Sol. $\Rightarrow 3xy + 10y = 2x + 5$

$y_1 = \frac{2-3y}{3x+10}, y_2 = \frac{-6(2-3y)}{(3x+10)^2}$

$y_3 = \frac{54(2-3y)}{(3x+10)^3}$

$\frac{y_1 y_3}{54} = \frac{{y_2}^2}{36}$

78. B

Sol. $f(x) = f(3x - 4y + 4y - 2x) = f(3x - 4y) + f(4y - 2x) - (3x - 4y)(4y - 2x)$

$f(x+y)=f(x)+f(y)-xy$

$f'(2)=\underset{h\to 0}{Lt}\dfrac{f(2+h)-f(2)}{h}=f'(0)$

79. B

Sol. $y=ke^{kx}$ cuts the y-axis at x = 0

$\dfrac{dy}{dx}=k^2e^{kx}$

at x = 0

$\dfrac{dy}{dx}=k^2$

$\alpha=\tan^{-1}(k^2)$ this is angle made by x-axis, angle made by y-axis $\dfrac{\pi}{2}-\tan^{-1}(k^2)$

or $\cot^{-1}(k^2)$

80. B

Sol. $T_1=\tan^{-1}\left(\dfrac{1}{1+x(1+x)}\right)=\tan^{-1}\left(\dfrac{1+x-x}{1+x(1+x)}\right)$

$=\tan^{-1}(1+x)-\tan^{-1}(x)$

$T_2=\tan^{-1}(2+x)-\tan^{-1}(1+x)$, etc

81. A

Sol. $S=\dfrac{16+3x+5x}{2}=8+4x$

Now $3x+5x>16\Rightarrow x>2$

$3x+16>5x\Rightarrow x<8\Rightarrow x\in(2,8)$

$5x+16>3x\Rightarrow x>-8$

Now $A^2(x)=(8+4x)(4x-18)(x+8)(8-x)=(16x^2-64)(64-x^2)$

Let $x^2=t\Rightarrow t\in(4,64), f(t)=16(t-4)(64-t)=16(64t-t^2-256+4t)$

$\therefore f(t)=-16(t^2-68t+256), f'(t)=2t-68=0\Rightarrow t=34$

$f''(t)=-32<0$

Maxima occurs at t=34

F(34)=16(30)(30)

$\therefore$ largest possible area = 120

82. D

Sol. Let $\int_0^t f(u)\,du=F(t)\Rightarrow F'(t)=f(t)$ and $F(0)=0$

Also $f(t)\ge 0\ \Rightarrow\int_0^t f(u)\,du\ge 0\ \forall\, t\in[0,1]$

$\Rightarrow |f(t)| \le \sqrt{1+2\int_0^t f(u)\,du}$

$\Rightarrow F'(t) \le \sqrt{1+2F(t)} \Rightarrow \dfrac{F'(t)}{\sqrt{1+2F(t)}} \le 1$ as $f(t) \ge 0$

$\int_0^t \dfrac{F'(t)\,dt}{\sqrt{1+2F(t)}} \le \int_0^t dt \;\forall\, t \in [0,1]$

$\Rightarrow \sqrt{1+2F(t)} \le t+1$

$\Rightarrow f(t) \le t+1$ Also $1+2F(t) \le (t+1)^2 \Rightarrow 2F(t) \le t^2+2t \Rightarrow F(t) \le \dfrac{t^2}{2}+t \Rightarrow \int_0^t f(u)\,du \le \dfrac{t^2}{2}+t$

83. B

Sol. $\int e^x \left(\dfrac{2\tan x}{1+\tan x} + \tan^2\left(x-\dfrac{\pi}{4}\right)\right)dx$

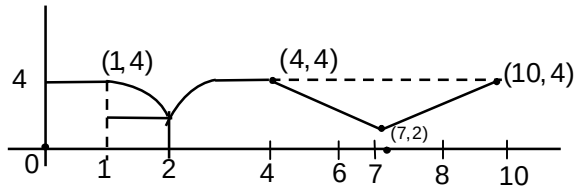
$= \int e^x \left(\tan\left(x-\dfrac{\pi}{4}\right) + \sec^2\left(x-\dfrac{\pi}{4}\right)\right)dx = e^x \tan\left(x-\dfrac{\pi}{4}\right)+c.$

84. C

Sol. Required area =

$2\left(4+\dfrac{\pi}{4}+3\right)+\left[(6\times 4)-\dfrac{1}{2}\times 2\times 6\right]$

$= 32+\dfrac{\pi}{2}$ sq. units



85. B

Sol. Probability of getting an odd number= $\dfrac{1}{9}$

Probability that getting an even number = $\dfrac{2}{9}$

Let 'A' denote the sum of the numbers thrown is even. $P(A) = 9\left(\dfrac{1}{81}\right)+9\left(\dfrac{4}{81}\right)=\dfrac{5}{9}$.

86. B

Sol. $\begin{vmatrix} 3 & a & bc \\ 3 & b & ac \\ 3 & c & ab \end{vmatrix} = \begin{vmatrix} 1 & a & 3bc \\ 1 & b & 3ac \\ 1 & c & 3ab \end{vmatrix}$

now $a^2-4bc=0,\ b^2-4ac=0,\ c^2-4ab=0$

$\Rightarrow a^2-bc=3bc,\ b^2-ac=3ac,\ c^2-ab=3ab$

$\Rightarrow \begin{vmatrix} 1 & a & a^2-bc \\ 1 & b & b^2-ac \\ 1 & c & c^2-ab \end{vmatrix} = 0$

$$\text{also}\begin{vmatrix}1 & a & \frac{3}{4}a^2\\ 1 & b & \frac{3}{4}b^2\\ 1 & c & \frac{3}{4}c^2\end{vmatrix}=\frac{3}{4}[a-b)(b-c)(c-a)$$

87. C

Sol Here $BC=\begin{bmatrix}1 & 4\\ 2 & 9\end{bmatrix}\begin{bmatrix}9 & -4\\ -2 & 1\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}=I$

The given matrix reduce to $A^3+A^5+A^7+.....+A^{2017}$ which is skew symmetric as A is skew symmetric

88. A

Sol. $\left|\vec{a}-2\vec{b}\right|^2+\left|\vec{b}-2\vec{c}\right|^2+\left|\vec{c}-2\vec{a}\right|^2=5a^2+5b^2+5c^2-4\left(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}\right)$

$=15-4\left(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}\right)\le 15-4\left(\frac{-3}{2}\right)=21$

89. B

Sol. $\left|\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]\right|=\frac{\sqrt{3}}{4}$

$$\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]=\begin{vmatrix}1+\sin\theta & \cos\theta & \sin 2\theta\\ \sin\left(\theta+\frac{2\pi}{3}\right) & \cos\left(\theta+\frac{2\pi}{3}\right) & \sin\left(2\theta+\frac{4\pi}{3}\right)\\ \sin\left(\theta-\frac{2\pi}{3}\right) & \cos\left(\theta-\frac{2\pi}{3}\right) & \sin\left(2\theta-\frac{4\pi}{3}\right)\end{vmatrix}$$

Apply $R_1\to R_1+R_2+R_3$.

$\left|\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]\right|=\frac{\sqrt{3}}{2}|\cos 3\theta|=\frac{\sqrt{3}}{4}$

90. C

Sol.

s	r	~r	~r∧s	~s	~s∨(~r∧s)	~(~s∨(~r∧s))
T	T	F	F	F	F	T
T	F	T	T	F	T	F
F	T	F	F	T	T	F
F	F	T	F	T	T	F

Which is the same as $s\wedge r$.